Database System Engineering and Distributed Backend Development

WEEK-2

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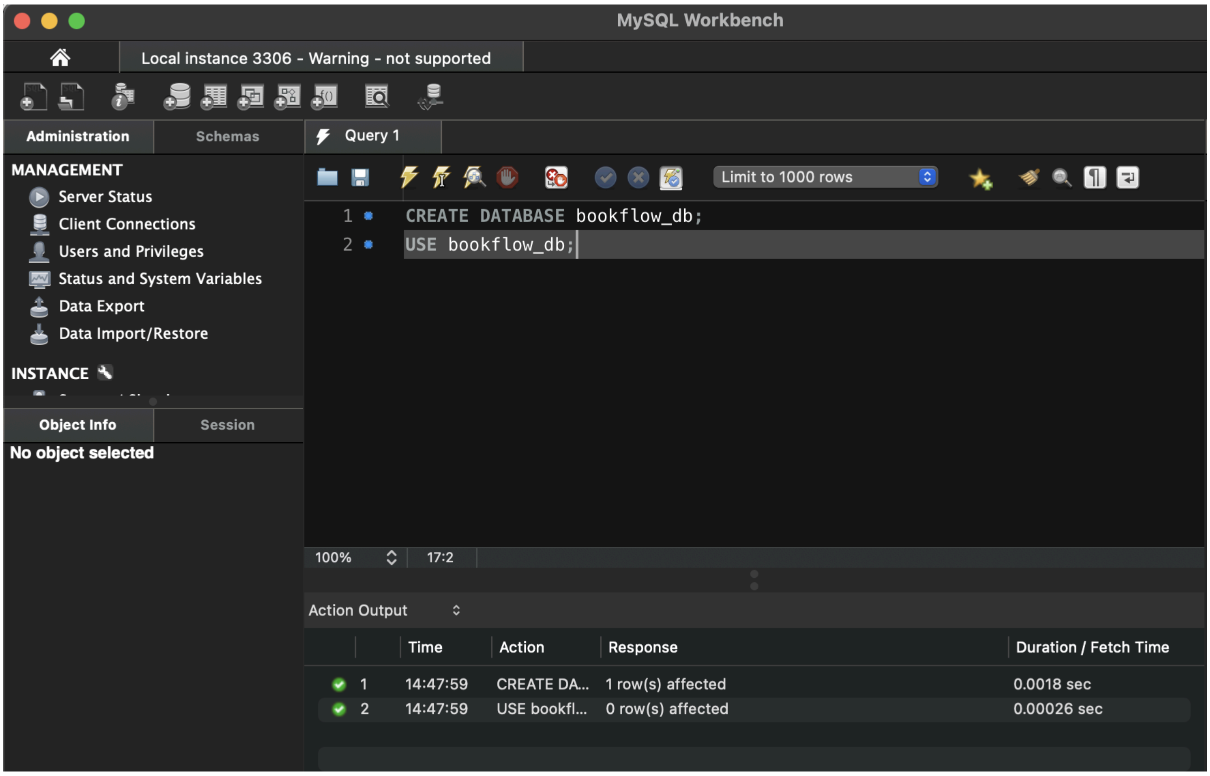
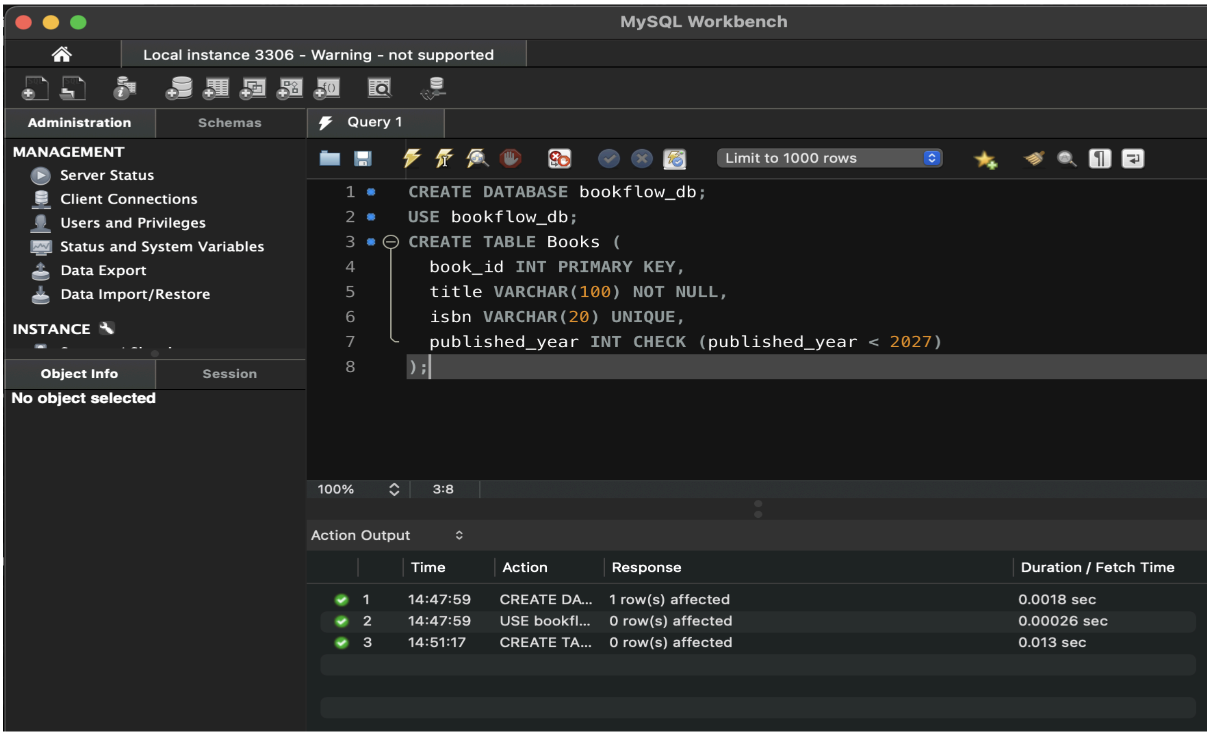
Step 1 — Create Database

```
CREATE DATABASE bookflow_db;
USE bookflow_db;
```



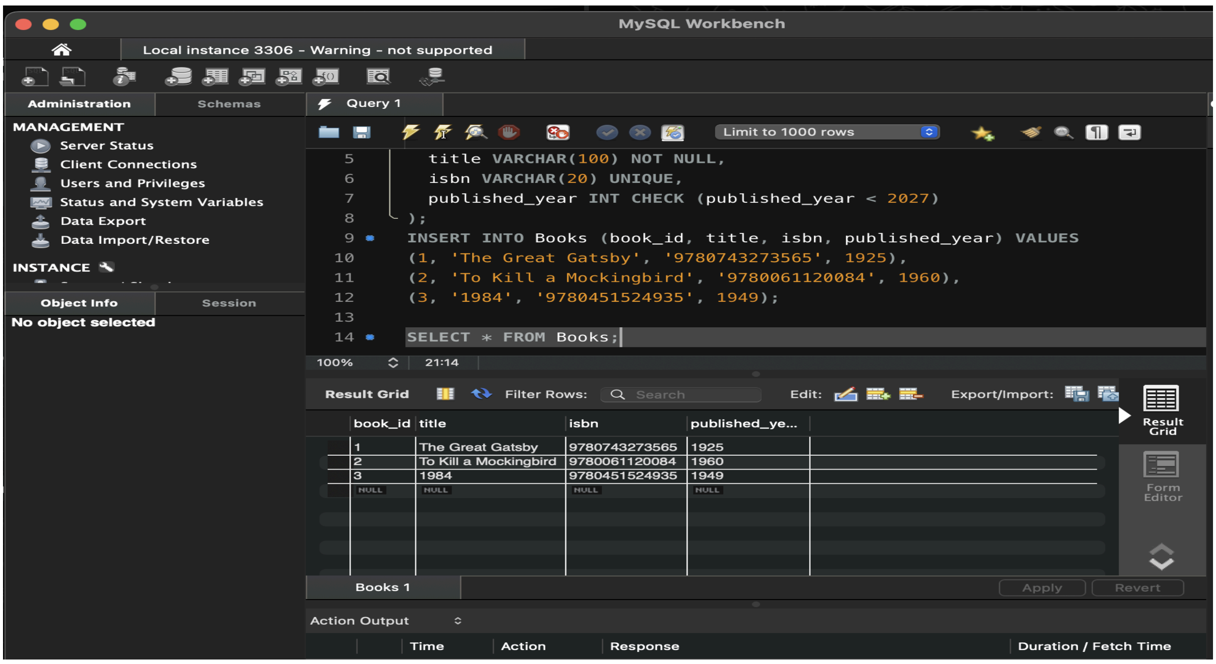
Step 2 — Create Books Table (DDL + Constraints)

```
CREATE TABLE Books (
  book_id INT PRIMARY KEY,
  title VARCHAR(100) NOT NULL,
  isbn VARCHAR(20) UNIQUE,
  published_year INT CHECK (published_year < 2027)
);
```



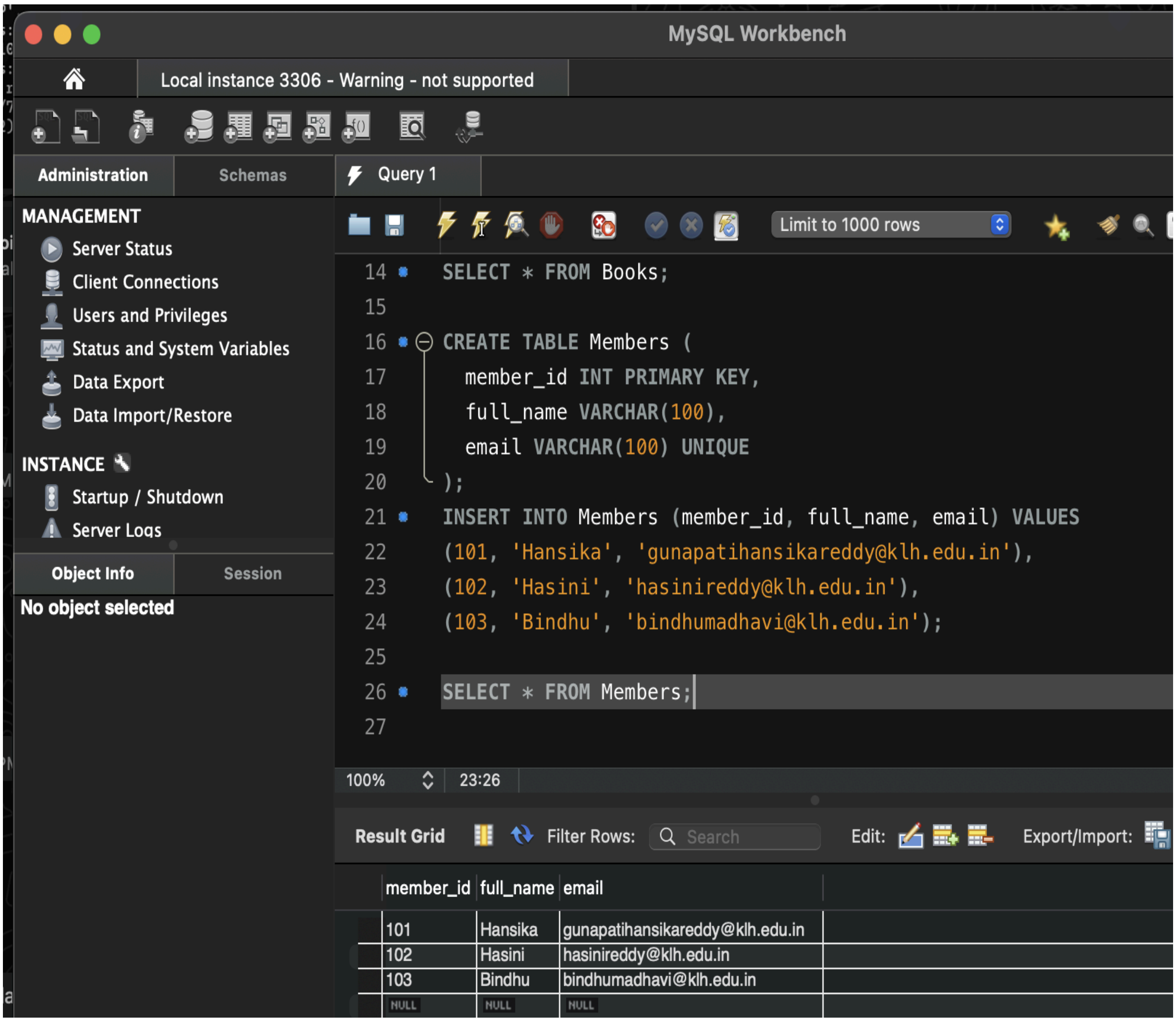
Step 3 & 4 — Insert Books and Display

```
INSERT INTO Books (book_id, title, isbn, published_year) VALUES
(1, 'The Great Gatsby', '9780743273565', 1925),
(2, 'To Kill a Mockingbird', '9780061120084', 1960),
(3, '1984', '9780451524935', 1949);

SELECT * FROM Books;
```



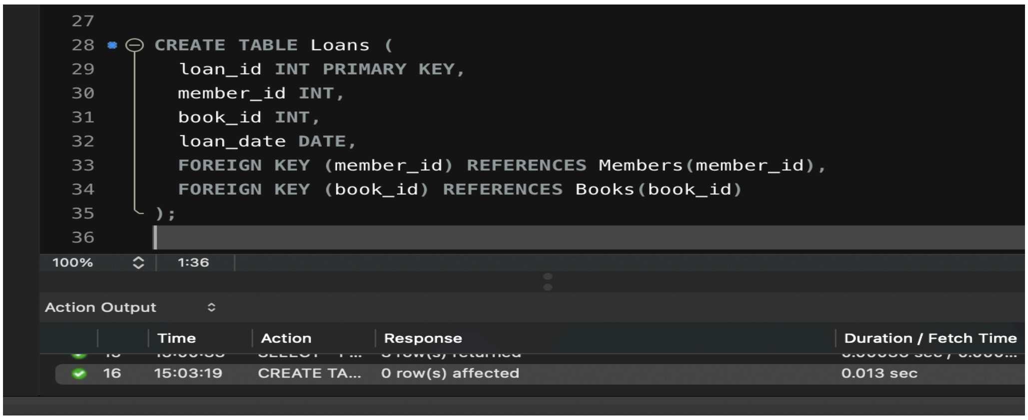
Step 5, 6 & 7 — Members Table: Create, Insert, Display

```
CREATE TABLE Members (
  member_id INT PRIMARY KEY,
  full_name VARCHAR(100),
  email VARCHAR(100) UNIQUE
);

INSERT INTO Members (member_id, full_name, email) VALUES
(101, 'John Smith', 'john.smith@email.com'),
(102, 'Emma Wilson', 'emma.wilson@email.com'),
(103, 'Michael Brown', 'michael.brown@email.com');

SELECT * FROM Members;
```



Step 8 — Create Loans Table (FOREIGN KEYS)

```
CREATE TABLE Loans (
  loan_id INT PRIMARY KEY,
  member_id INT,
  book_id INT,
  loan_date DATE,
  FOREIGN KEY (member_id) REFERENCES Members(member_id),
  FOREIGN KEY (book_id) REFERENCES Books(book_id)
);
```

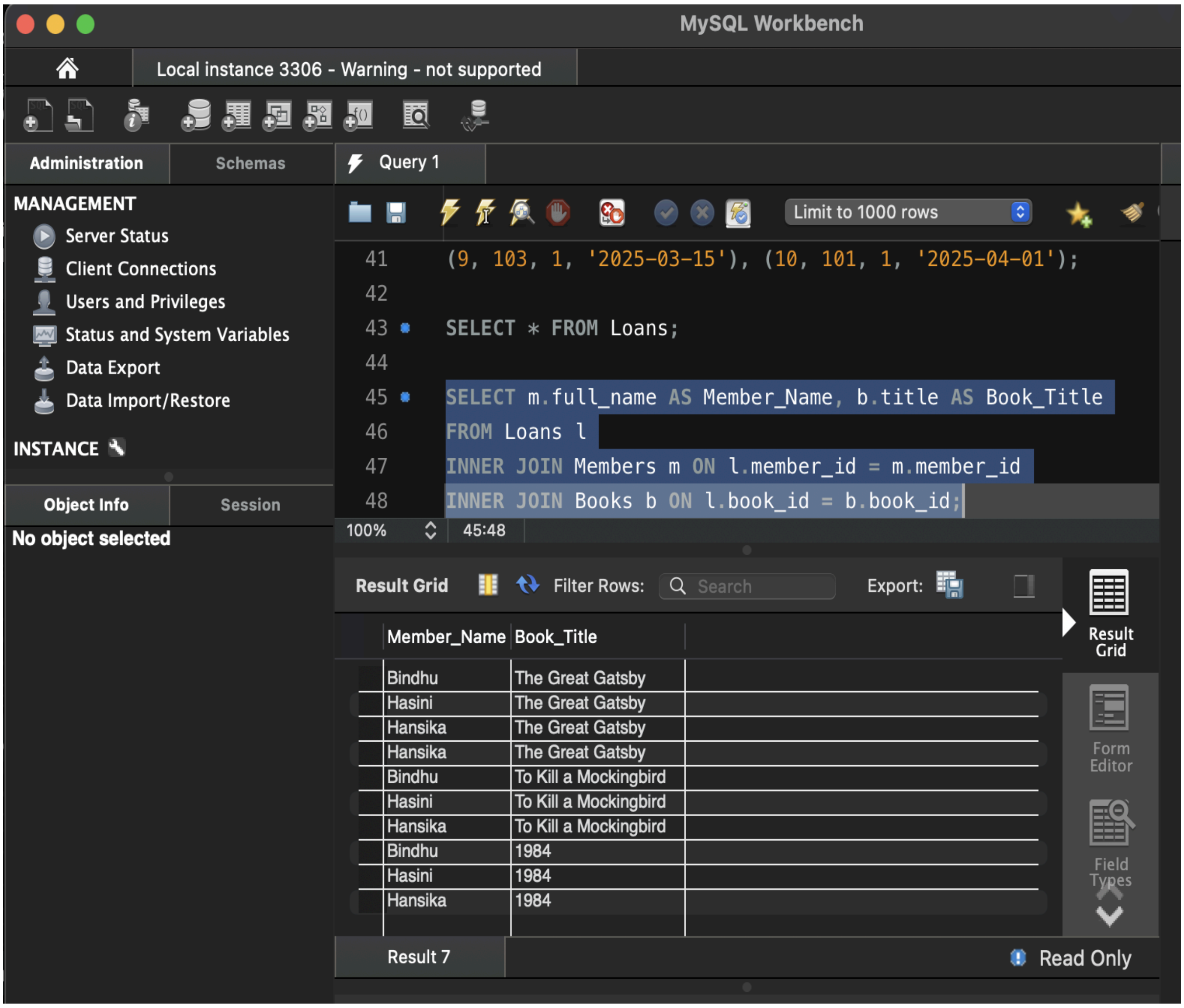


Step 9 & 10 — Insert 10 Loans and Display

```
INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
(1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
(3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
(5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
(7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
(9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');

SELECT * FROM Loans;
```

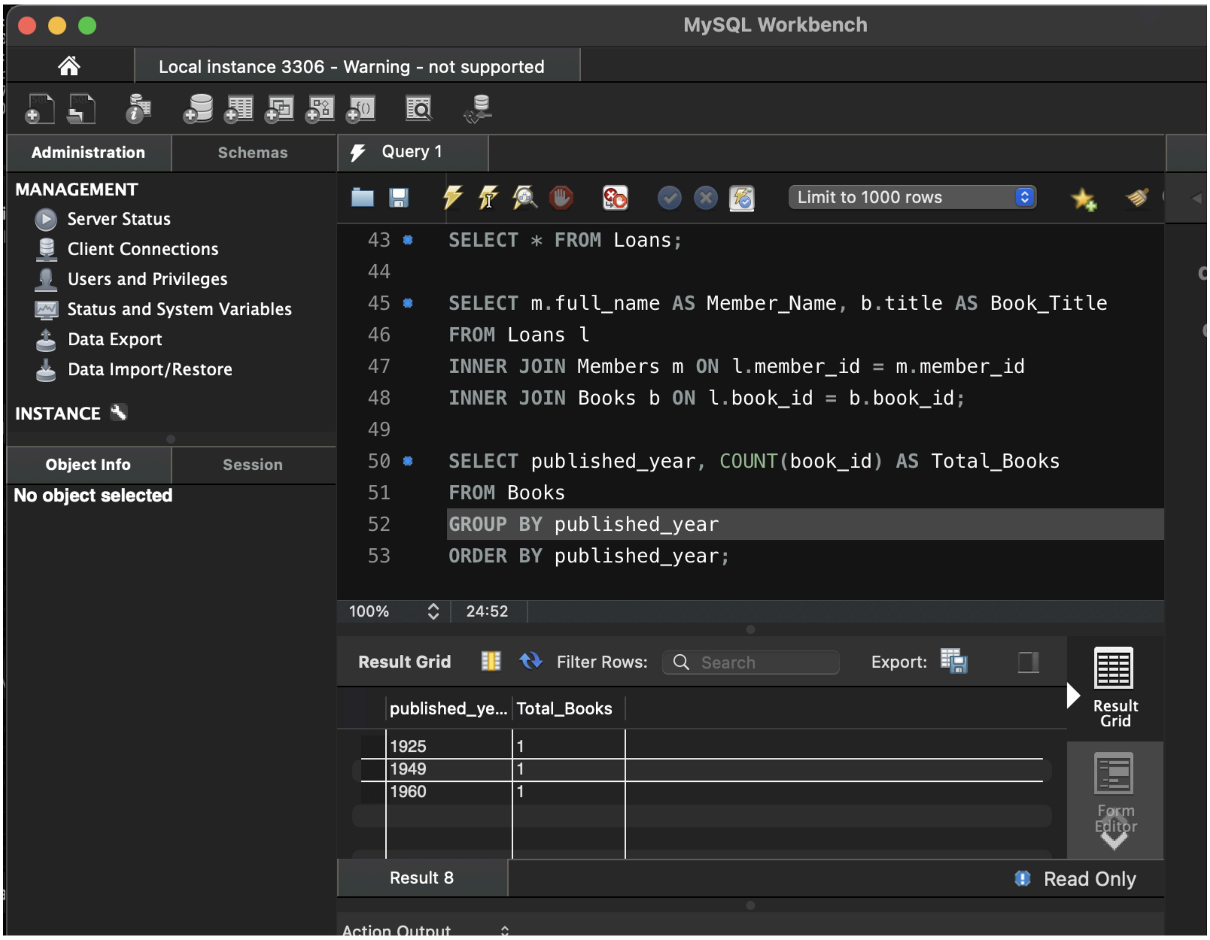
Step 11 — JOIN Query (Who Borrowed What)

```
SELECT m.full_name AS Member_Name, b.title AS Book_Title
FROM Loans l
INNER JOIN Members m ON l.member_id = m.member_id
INNER JOIN Books b ON l.book_id = b.book_id;
```



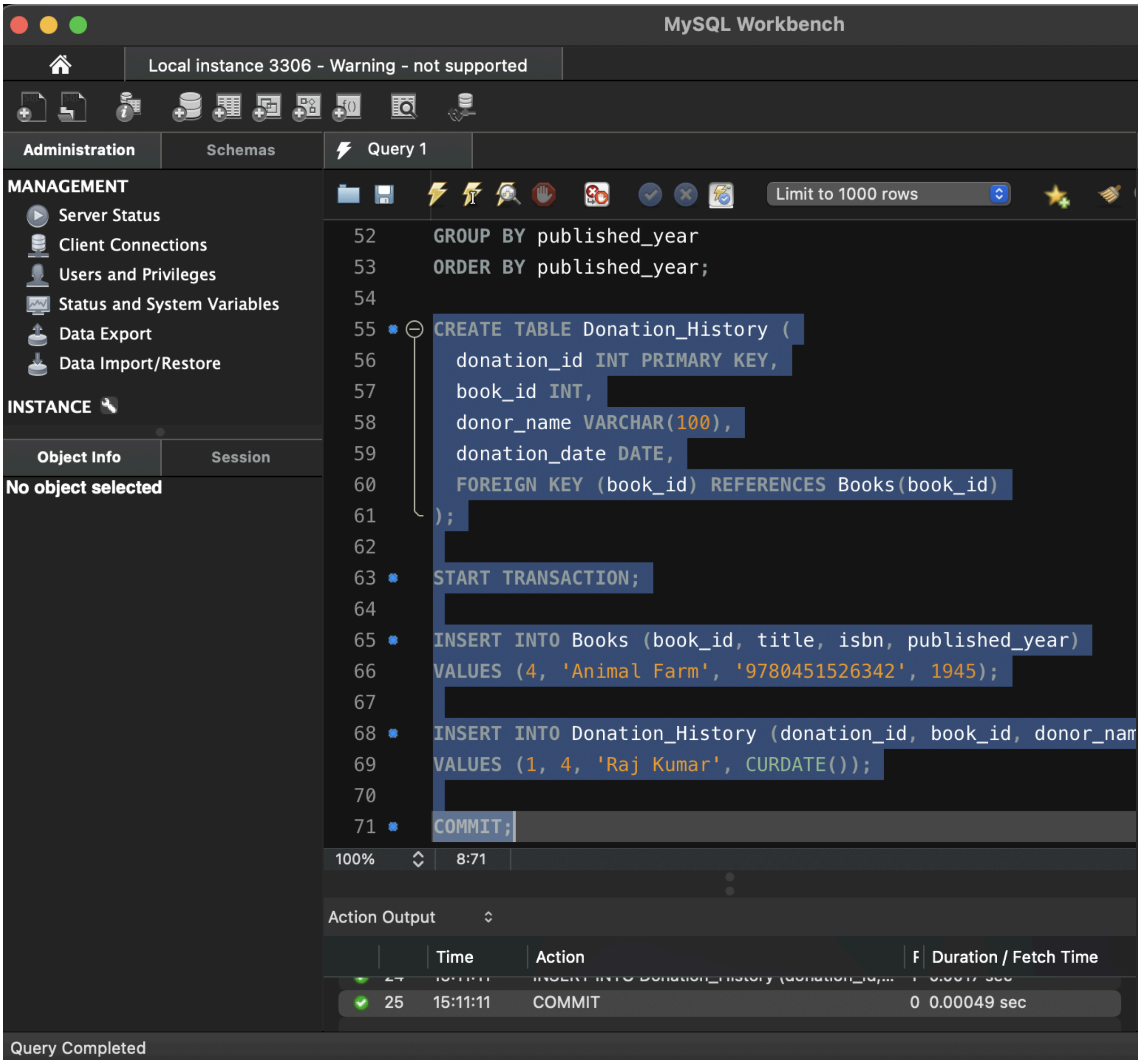
Step 12 — GROUP BY Query (Books per Year)

```
SELECT published_year, COUNT(book_id) AS Total_Books
FROM Books
GROUP BY published_year
ORDER BY published_year;
```



Step 13–16 — Donation via TRANSACTION (Atomicity)

```
CREATE TABLE Donation_History (
  donation_id INT PRIMARY KEY,
  book_id INT,
  donor_name VARCHAR(100),
  donation_date DATE,
  FOREIGN KEY (book_id) REFERENCES Books(book_id)
);

START TRANSACTION;

INSERT INTO Books (book_id, title, isbn, published_year)
VALUES (4, 'Animal Farm', '9780451526342', 1945);

INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
VALUES (1, 4, 'Raj Kumar', CURDATE());

COMMIT;
```



Step 17 & 18 — Index on ISBN + Fast Search

```
CREATE INDEX idx_books_isbn ON Books(isbn);

SELECT * FROM Books WHERE isbn = '9780451524935';
```